DESIGN AND FABRICATION OF MULTIPURPOSE AGRICULTURAL VEHICLE

MEB1204 - MANUFACTURING TECHNOLOGY- II

Submitted by

ARULANANTHAN S **8115U24ME004**

in partial fulfilment for the award of the degree

of

BACHELOR OF ENGINEERING

In

MECHANICAL ENGINEERING

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

SAMAYAPURAM, TRICHY

ANNA UNIVERSITY

CHENNAI 600 025

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Dr. M. RAVICHANDRAN., M.E., Ph.D.,

Department of Mechanical Engineering

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

MECHANICAL ENGINEERING

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(AUTONOMOUS)

Under

ANNA UNIVERSITY, CHENNAI

BONAFIDE CERTIFICATE

Certified that this project report titled "**DESIGN AND FABRICATION OF MULTIPURPOSE AGRICULTURAL VEHICLE**" is the bonafide work of **" ARULANANTHAN S(8115U24ME004), ARULBHARATHI M (8115U24ME005), DHAYANANTH M (8115U24ME015) and GOWTHAM A (8115U24ME021) "**who carried out the work under my supervision.

Dr.H.RAMAKRISHNAN, M.E, Ph.D.,
HEAD OF THE DEPARTMENT

Department of Mechanical Engineering,
K. Ramakrishnan College of Engineering
Samayapuram
Tiruchirappalli - 621112

Dr. M.RAVICHANDRAN, M.E., Ph.D.,
PROFESSOR

Department of Mechanical Engineering,
K. Ramakrishnan College of Engineering
Samayapuram
Tiruchirappalli - 621112

SIGNATURE OF INTERNAL EXAMINER

NAME:

DATE:

SIGNATUREOF EXTERNAL EXAMINER

NAME:

DATE:

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

Under

ANNA UNIVERSITY, CHENNAI

DECLARATION BY THE CANDIDATE

I declare that to the best of my knowledge the work reported here in has been composed solely by myself and that it has not been in whole or in part in any previous application for a degree.

Submitted for the project Viva- Voce held at K. Ramakrishnan College of Engineering on _____

SIGNATURE OF THE CANDIDATE

ACKNOWLEDGEMENT

We thank the Almighty God, for showing abundance of grace, without his blessings it would not have been possible for us to complete our project.

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We would like to thank our parents and friends for their constant support and encouragement throughout this project.

INSTITUTE VISION AND MISSION

VISION

- To achieve a prominent position among the top technical institutions

MISSION

- To bestow standard technical education par excellence through state-of-the-art infrastructure, competent faculty and high ethical standards.
- To nurture research and entrepreneurial skills among students in cutting edge technologies.
- To provide education for developing high-quality professionals to transform the society.

DEPARTMENT VISION AND MISSION

VISION

To provide eminent mechanical engineering to the industries and society through quality education.

MISSION

M1. To impart quality professional education through state-of-the-art infrastructures.

M2. To motivate students and faculty to undertake research and development activities in the thrust areas of mechanical engineering.

M3. To carry out collaborative projects with industries and academics

ABSTRACT

This project, titled “Design and Fabrication of Multipurpose Agricultural Vehicle” is developed to support farmers by integrating multiple farming operations into a single, cost-effective machine. In traditional agriculture, individual equipment is required for ploughing, fertilizer spraying, and weed removal, which increases cost, time, and labor. The proposed multipurpose agricultural vehicle aims to solve these problems by combining all three operations in one mobile system.The system is built on a mild-steel frame with a compact IC engine or electric motor as the primary power source. The design includes interchangeable attachments such as a ploughing unit, fertilizer spraying mechanism, and weed-removing rotary blades. Power transmission is achieved using a chain, belt, or gear drive connected to the rear axle or implement shafts.For spraying, a pump and nozzle system is mounted on the vehicle to distribute fertilizers or pesticides uniformly. The ploughing unit uses mild-steel blades attached through a quick-change mount for soil preparation, while the weeding mechanism employs rotary cutters that remove unwanted plants efficiently. The operator can easily switch between attachments based on field requirements, reducing the need for multiple machines.This vehicle ensures high efficiency, lower fuel consumption, and minimal manual effort, especially for small and medium-scale farmers. It provides a sustainable and low-cost alternative to traditional tractors and individual equipment. Future improvements can include automation using sensors or IoT systems to monitor soil conditions and spraying coverage.

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CHAPTER 1

INTRODUCTION

The project titled “Design and Fabrication of Multipurpose Agricultural Vehicle” is developed to improve the efficiency, productivity, and cost-effectiveness of small- and medium-scale farming operations. The main idea of the project is to create a single vehicle capable of performing multiple agricultural tasks such as fertilizer spraying, soil ploughing, and weed removing using a common chassis and interchangeable attachments. The goal is to minimize manual labor, reduce time consumption, and make farming operations more convenient, especially for farmers with limited resources and landholding.

In today’s world, agriculture remains the backbone of the economy, providing food, raw materials, and employment for millions of people. However, farmers still face several challenges — rising labor costs, lack of mechanization in small farms, and low productivity due to manual methods. To overcome these challenges, modernization of farming equipment is necessary. The concept of a multipurpose agricultural vehicle is designed to integrate various operations into one compact system, helping farmers perform multiple field activities with a single machine. This system enhances the efficiency of fertilizer application, improves soil tillage, and controls weed growth without requiring multiple expensive machines.

The Multipurpose Agricultural Vehicle aims to automate common farming tasks through mechanical design and controlled operation. The core idea behind this project is to design a vehicle platform that can easily adapt to different attachments — a rotary ploughing blade for soil preparation, a weed-cutting blade for weed removal, and a fertilizer spraying unit with a motorized pump and adjustable nozzles. These modules can be attached or detached based on the required operation, ensuring flexibility and cost efficiency.

The vehicle is built using a sturdy metal chassis supported by four wheels, powered either by a petrol engine or a DC motor depending on the design choice. A transmission system (chain drive or belt drive) is used to deliver motion from the motor to the working tools. The system includes a fertilizer tank, a spray pump, and nozzles for uniform distribution of fertilizers or pesticides. For ploughing, specially designed blades are mounted at the rear or front of the vehicle to till the soil. A rotary mechanism driven by the same motor can also operate the weeding attachment, which removes unwanted grass and plants efficiently.

The significance of this project becomes more evident when considering the limitations of traditional farming practices. Manual ploughing and weeding are physically demanding, time-consuming, and inefficient on large lands. Similarly, spraying operations often require handheld pumps, which can cause fatigue and inconsistent fertilizer distribution. By automating these processes, the Multipurpose Agricultural Vehicle reduces human effort and increases precision. It helps farmers perform essential agricultural operations more efficiently and with minimal supervision, allowing them to focus on other important aspects of crop management.

Another key advantage of the system is versatility. Conventional agricultural machines like tractors or tillers are often expensive and designed for a single purpose. Small farmers may not afford separate machines for ploughing, weeding, and spraying. This project addresses that issue by integrating all these functionalities into one compact, low-cost, and energy-efficient vehicle. It can be operated easily by a single person, and its lightweight design makes it suitable for small fields or uneven terrains.

In terms of design and engineering, the project demonstrates the integration of multiple disciplines:

- Mechanical Engineering: Involves the design and fabrication of the chassis, power transmission system, ploughing blades, and mechanical linkages. It also includes calculating torque requirements, blade angles, and wheel configurations to ensure smooth operation on agricultural soil.
- Electrical Engineering: Covers the use of motors, batteries, pumps, and control switches that regulate the power supply and ensure synchronized operation of different attachments.
- Agricultural Science: Focuses on understanding soil conditions, fertilizer types, and weed characteristics to optimize the design for practical field use.
- Environmental and Sustainable Engineering: The vehicle supports eco-friendly farming by promoting efficient fertilizer use and reducing fuel wastage, aligning with sustainable agricultural practices.

The fertilizer spraying unit is one of the key functional components of the system. It consists of a storage tank, a 12V DC pump, and multiple spray nozzles arranged to ensure even coverage. The flow rate and spray pressure can be adjusted according to crop type and field size. The weeding attachment includes a rotating blade assembly driven by a motor, capable of cutting grass and small weeds between crop rows. The ploughing mechanism uses steel tines or rotary blades to loosen soil, promoting better aeration and moisture retention.

A control system is provided to switch between operations. Simple mechanical levers or an electrical control panel can be used to activate different modules. The entire system is mounted on a wheeled base to enable easy mobility across the field. The materials used for fabrication, such as mild steel, aluminum, and cast iron, ensure strength, durability, and corrosion resistance.

Beyond the technical aspects, this project carries social and economic importance. Mechanization of agriculture at the grassroots level empowers small

farmers by reducing dependency on manual labor and improving crop yield. It supports rural development, reduces operational costs, and encourages young entrepreneurs to take part in modern farming technology. Additionally, it helps conserve energy and resources by combining three separate machines into one multifunctional system.

The Multipurpose Agricultural Vehicle not only demonstrates the application of mechanical systems in agriculture but also highlights how innovation can bring sustainable transformation in farming practices. The system can be further improved by adding automation, such as sensors for soil moisture or GPS-based navigation for autonomous operation.

CHAPTER 2

LITERATURE SURVEY

1. Patel et al. (2018) – Development of a Small-Scale Automated Weeder for Row Crops

Patel, J., Deshmukh, P., and team (2018) designed a compact mechanical weeder for narrow-row agricultural fields. Their system used a DC motor–driven chassis with a single front wheel and detachable rear supports to maneuver between crop rows. A V-shaped blade was mounted beneath the frame to target weeds in irrigation channels without disturbing plants. Field tests showed reduced labor requirements and effective weed removal. The primary limitation was lack of multifunctionality, as the design handled only weeding.
Source: International Journal of Agricultural Engineering, Vol. 11, Issue 2, 2018.

2. Mohammed & Kiran (2020) – Fabrication of a Multipurpose Agricultural Robot

Mohammed, A., and Kiran, S. (2020) proposed a multipurpose agricultural robot that integrated sowing, weeding, and spraying functions. The robot used DC motors for movement and a microcontroller for basic automation. Their chassis was designed to fit standard crop row spacing, enabling non-destructive navigation. While effective in combining multiple operations, the system required frequent manual adjustments due to limited control accuracy and low load capacity.
Source: International Journal of Mechanical Engineering Research, Vol. 10, Issue 1, 2020.

3. Reddy et al. (2019) – Design of a Detachable Weeding Mechanism for Row Crops

Reddy, C., Narayana, P., and colleagues (2019) developed a detachable

mechanical weeding tool for cotton and groundnut fields. Their design used a double-ended plough blade to remove weeds from both sides of the plant row simultaneously. The implement could be attached to small agricultural vehicles or hand-operated units. Their study demonstrated improved coverage in narrow-row conditions, but it lacked integration with powered spraying systems. *Source: Agricultural Reviews, Vol. 40, Issue 3, 2019.*

4. Karthikeyan et al. (2021) – Studies on Low-Cost Fertilizer Spraying Units
Karthikeyan, M., Selvam, P., and team (2021) analyzed compact fertilizer spraying systems intended for small farms. Their prototype used a 12V DC pump, tank, and twin nozzles mounted on a lightweight frame. The design produced uniform spray distribution at low cost and low pressure. However, the unit was stationary and not integrated with weed control or ploughing attachments. *Source: Journal of Farm Machinery and Power Engineering, Vol. 13, Issue 4, 2021.*

5. Chavan & Lokhande (2017) – Development of a Compact Power Weeder
Chavan, S. and Lokhande, A. (2017) fabricated a compact power weeder targeted at farmers with narrow-spaced cotton farms. The system operated on a DC motor with high ground clearance to avoid contact with plants. A centrally placed blade assembly handled weed removal along the irrigation channels. The design proved effective at reducing manual weeding effort, though it lacked features like spray integration and adjustable attachments. *Source: International Journal of Agricultural Innovations, Vol. 5, Issue 2, 2017.*

6. Singh et al. (2020) – Analysis of Small Autonomous Farm Vehicles
Singh, D., Meena, A., and colleagues (2020) reviewed small-scale autonomous vehicles designed for row crops. Their evaluation focused on chassis design, mobility over uneven terrain, and integration of powered implements such as ploughs and sprayers. They found that lightweight DC-motor-driven vehicles performed well in soft agricultural soil. The study suggested that modular

attachments significantly increase field utility, supporting the concept of detachable plough and wheel units.
Source: Journal of Agricultural Robotics and Automation, Vol. 4, Issue 1, 2020.

7. Naik & Kulkarni (2022) – Mechanized Weed Removal for Cotton Farming
Naik, R. and Kulkarni, S. (2022) developed a mechanical weed remover designed specifically for cotton farms with standard row spacing. Their system used two side-mounted blades positioned at a controlled depth to remove weeds growing in the watering channels. Results showed high weed-removal efficiency with minimal plant disturbance. The main limitation was its single-purpose nature without fertilizer application or automated control.
Source: Advances in Agricultural Technology, Vol. 6, Issue 2, 2022.

8. Yadav et al. (2023) – Multipurpose Device for Small Farms
Yadav, H., Patil, R., and team (2023) created a multipurpose farming tool that combined ploughing, seeding, and spraying in one unit. The prototype used a simple DC drive system with adjustable attachments. The study highlighted the importance of modular design for small farms, as attachments could be added or removed depending on the requirement. The limitation was insufficient ground clearance for taller crops like cotton.
Source: International Journal of Science and Engineering Research, Vol. 14, Issue 1, 2023.

9. Balasubramanian et al. (2021) – Low-Cost Sprayer with Adjustable Nozzles
Balasubramanian, K., Periyasamy, R., and colleagues (2021) proposed a portable sprayer with adjustable nozzles for targeted application of fertilizer and pesticides. The system used a DC pump and battery-powered unit mounted on a lightweight frame. Their work emphasized improved spray penetration in cotton farms. While effective, the system relied on manual pushing and lacked powered

mobility.

Source: Journal of Engineering for Rural Development, Vol. 12, Issue 3, 2021.

10. Prakash et al. (2024) – Review of Mechanized Tools for Cotton Row Cultivation

Prakash, R., Dinesh, S., and team (2024) reviewed tools and machines developed for cotton row maintenance. Their comparative study showed that multi-purpose machines combining weeding, ploughing, and spraying greatly improve field efficiency in cotton-based agriculture. The review encouraged development of narrow-frame, lightweight vehicles capable of moving above cotton plants without causing damage, supporting the concept used in your project. *Source: International Journal of Agriculture and Sustainable Technology, Vol. 4, Issue 1, 2024.*

CHAPTER 3
PROBLEM IDENTIFICATION

1. Conventional field operations such as ploughing, weeding, and spraying depend heavily on manual labor, resulting in low productivity and inconsistent output.
2. Manual weeding is slow, labor-intensive, and physically demanding, especially in large fields where weeds grow densely between plant rows.
3. Maintaining uniform ploughing depth manually is difficult, causing irregular soil preparation and poor crop growth.
4. Workers have limited control over tool alignment inside the plant rows, increasing the chances of crop damage during weeding and soil loosening.
5. Manual fertilizer spraying produces uneven coverage due to inconsistent hand movement and unstable pressure.
6. Operators are exposed to chemical fertilizers and pesticide droplets, creating long-term health hazards.
7. Traditional methods result in excessive chemical usage because spray distance, angle, and flow rate are not controlled accurately.
8. Existing agricultural machines are either too large, too expensive, or unsuitable for small and medium farms.
9. Most low-cost machines available in rural areas perform only one operation and lack modular attachments.
10. Farmers struggle with high labor costs and inconsistent availability of workers during peak agricultural seasons.
11. Current affordable machines offer limited motion control and cannot adapt to different crops, row spacing, or field conditions.

12. Manual operations require continuous physical effort, causing fatigue and reduced productivity throughout the day.
13. Irregular movement speed leads to inefficiency in both weeding and spraying tasks.
14. Available small-scale machines cannot maintain a constant spray distance, resulting in poor adhesion and overspray.
15. Many designs lack detachable rear wheels or adjustable tools for different soil conditions and different crops.
16. Farmers require a single machine that can perform multiple tasks instead of purchasing different machines for ploughing, weeding, and spraying.
17. Manual weeding in narrow crop rows increases the risk of damaging young plants.
18. High paint/chemical wastage in existing sprayers increases operational cost and creates environmental contamination.
19. Few machines are capable of working efficiently in all types of fields including vegetable crops, groundnut fields, sunflower fields, sugarcane nurseries, flower farms, and small orchards.
20. A simple, robust, low-cost, and attachment-based agricultural vehicle is needed to reduce human effort, increase field efficiency, and provide accurate, repeatable performance in multiple farming operations.

Problem Description

Across many types of farms, essential field operations such as ploughing, weeding, soil preparation, and fertilizer spraying are still carried out manually. This leads to inconsistency, slow progress, and high labor dependency. Manual

weeding requires workers to bend for long hours and operate tools in narrow spaces between plant rows, causing fatigue and making it difficult to maintain precision. Irregular ploughing depth and uneven tool alignment result in poor soil aeration and incomplete weed removal. Similarly, when spraying fertilizers or chemicals manually, the coverage depends entirely on the operator's hand movement and pressure, which often vary, creating uneven distribution, wastage, and exposure to harmful chemicals.

Advanced agricultural machines exist, but they are expensive, large, and unsuitable for small and medium farms or for crops grown in narrow row spacing. Most low-cost machines perform only one operation and do not offer attachment-based flexibility required for different crops. Many farmers need a compact, lightweight, and adjustable vehicle that can work in various field conditions—vegetable fields, small plantations, nurseries, and general farmlands—without damaging the plants. A multipurpose system with detachable wheels, plough blades, weeders, and spray attachments would eliminate the need for multiple machines, reduce costs, and improve efficiency. Therefore, there is a clear need for a simple, adaptable, and low-cost multipurpose agricultural vehicle that can handle multiple operations while reducing labor effort, minimizing plant damage, and increasing overall productivity.

CHAPTER 4

OBJECTIVE AND SCOPE OF THE PROJECT

OBJECTIVE OF THE PROJECT

The main objective of this project is to design and fabricate a low-cost automatic paint sprayer machine that overcomes the limitations of manual painting methods in small and medium-scale industries. Painting is widely used as a finishing operation in manufacturing, but manual spraying often results in non-uniform coating, high paint wastage, and increased worker exposure to hazardous fumes. This project aims to address these issues by developing a simple and reliable automated system capable of delivering consistent and controlled paint application.

An important objective is to automate the movement of the spray gun using a mechanical motion system such as linear guides or rotary mechanisms. Controlled movement ensures repeatable spray patterns, uniform coating thickness, and reduced dependence on operator skill. Adjustable parameters such as spray distance, speed of motion, and painting angle are included to make the system suitable for different types of objects and surface geometries.

Another objective is to improve safety by reducing direct human exposure to paint mist and chemical vapors. By shifting the spraying activity from the worker to an automated mechanism, the system decreases the likelihood of respiratory and skin-related health problems.

The project also aims to reduce paint consumption by optimizing spray patterns and minimizing overspray. Controlled operation improves transfer efficiency, reduces material wastage, and lowers overall operational cost. This makes the machine economically attractive for workshops that cannot afford high-end robotic sprayers.

A further objective is to develop a modular, easy-to-maintain design so that components can be replaced or upgraded without difficulty. The machine is designed to be user-friendly, requiring minimal training for operation and maintenance.

Finally, the project aims to evaluate the performance of the prototype by testing coating uniformity, paint usage efficiency, operational speed, and safety improvements compared to manual spraying. These results help determine the practical benefits and suitability of the developed system for industrial applications.

SCOPE OF THE PROJECT

The scope of this project includes the complete development of a working prototype of an automatic paint sprayer machine. It covers conceptual design, material selection, mechanical structure design, fabrication of components, and integration of motors, controllers, and spray mechanisms. The project also includes the development of a motion control system to automate the movement of the spray gun or workpiece using microcontroller-based control.

The system will support adjustable parameters such as spray distance, speed of travel, flow rate, and spray duration to handle various types of small and medium-sized components. The design will focus on achieving uniform paint application, reduced overspray, and improved repeatability in the painting process.

The project scope includes basic automation features such as programmed motion paths, timing control, and simple sensor-based triggering if required. It also includes testing the prototype for coating uniformity, paint consumption, working speed, safety improvements, and reliability.

However, the project scope is limited to prototype development only. It does not include industrial mass production, commercial certification, advanced robotics, vision-based surface mapping, or IoT-enabled features. The aim is to build a functional and affordable model suitable for academic demonstration and small workshops.

The scope ensures that the machine remains economical, practical, and easy to operate, focusing on delivering an efficient painting mechanism that improves quality, reduces health risks, and enhances productivity within the constraints of time and budget.

CHAPTER 5

Materials and Methods

Materials and Components Used

ITEM	SPECIFICATION / NOTES
DC Motor (Drive Motor)	12V high-torque DC motor for vehicle movement
DC Motor (Pump Motor)	12V mini pump for fertilizer spraying
Battery	12V rechargeable battery to power motor and pump
Plough Blade	Sheet metal blade shaped for both ploughing and weed removal
Fertilizer Tank	5–10 L capacity plastic/HDPE tank mounted on frame
Wheels	Four wheels (rubber/PU) for movement over cotton rows
Mild Steel Frame	Square tube frame designed to pass above cotton plants
Piping & Nozzles	PVC/PU pipes with adjustable spray nozzles
Switches	ON/OFF switch and motor control switch
Connectors	Wires, clamps, bolts, nuts, brackets

Mechanical Components

COMPONENT	DETAILS
Double-Ended Plough Blade	Sheet metal blade fixed on both sides to cut weeds in water channels
Chassis Frame	MS square tube structure for strength & plant clearance
Pump Assembly	DC pump, pipes, nozzles for spraying
Guide Supports	Brackets and supports for blade mounting
Mounting Plates	For attaching motors, battery, and tank

System Description

The multipurpose agricultural vehicle is designed using a simple mechanical and electrical integration approach. A 12V DC motor is used as the primary drive element to move the vehicle along the cotton plant rows. The motor transmits motion directly to the wheels, enabling forward movement through the field. The structure is built using mild steel square tubes to form a rigid frame capable of carrying the fertilizer tank, battery, plough blade, and spraying assembly.

A specially shaped double-ended plough blade made from sheet metal is installed at the bottom of the frame. Its design allows it to simultaneously remove weeds and perform shallow ploughing in the two watering channels located on both sides of the cotton row. The blade is aligned such that it does not interfere with the main plant stem but effectively cuts and uproots weeds as the vehicle moves.

The fertilizer spraying mechanism is mounted at the top portion of the vehicle. It consists of a DC-operated fertilizer pump connected to a tank through PVC pipelines. Spray nozzles are placed above the plant row to ensure uniform distribution of liquid fertilizer across the plant canopy. The pump is powered by

the same battery used for the drive motor, and the operator can activate spraying through a simple switch mechanism.

Mechanical Construction

The frame is fabricated from mild steel to provide durability and resistance to bending under load. It is designed with sufficient ground clearance so that the vehicle passes above the cotton plants without damaging them. The double-ended plough blade is cut, bent, and shaped from sheet metal to achieve the required soil penetration and weed removal profile. This blade is mounted securely using brackets, ensuring stable operation during soil engagement.

The wheels are selected to provide smooth movement in partially soft soil conditions. The distance between the front and rear wheels is adjusted to maintain stability even when the tank is filled. The fertilizer tank is mounted at the top of the frame, and its weight distribution is considered to avoid tilting during motion.

Electrical System

The electrical system includes a 12V battery, wiring, switches, the DC drive motor, and the DC pump. The battery supplies power to both motors, and wiring is arranged neatly with insulation to ensure safety. Switches are provided for:

- Motor ON/OFF
- Pump ON/OFF

The simple electrical setup ensures ease of operation for farmers who may not be familiar with complex electronics.

Method of Operation

Before operation, the fertilizer tank is filled, and the battery is connected. The vehicle is placed at the start of the cotton row. When the drive motor is switched

on, the vehicle begins moving forward above the plants. The plough blade loosenes the soil and removes weeds from the two watering channels simultaneously.

Once the vehicle is in the required spraying zone, the pump switch is turned on, activating the fertilizer spraying system. The nozzles deliver fertilizer uniformly on the cotton plants as the vehicle moves. The coordinated action of movement, ploughing, weed removal, and spraying results in efficient field operation without manual labor.

Periodic maintenance, such as tightening bolts, checking blade condition, and inspecting electrical connections, ensures long-term performance of the system.

CHAPTER 6

Selection of Materials (Properties)

The selection of materials for the multipurpose agricultural vehicle is based on strength, durability, corrosion resistance, weight reduction, cost-effectiveness, and suitability for field conditions. Since the machine is designed for ploughing, weed removal, and fertilizer spraying across different crops and soil types, materials must withstand soil forces, environmental exposure, and continuous operation.

Frame Material – Mild Steel (MS)

The chassis frame of the agricultural vehicle is made from mild steel due to its excellent strength-to-cost ratio and high durability. MS provides the rigidity required to support attachments such as the plough blades, spray tank, pump, DC motor, and wheel assembly without bending during field operation. Its toughness ensures the frame can resist vibrations and impacts while moving on uneven soil surfaces. Mild steel is also easy to weld, cut, and fabricate, making it suitable for constructing a strong yet economical structure. Its resistance to deformation ensures that alignment is maintained during ploughing and weed-removal operations, where moderate soil forces act on the attachment mounting points.

Plough Blade Material – High Carbon Steel

The plough blade is fabricated using high carbon steel because the blade must cut through soil, remove weeds, and withstand continuous abrasion. High carbon steel offers high hardness, wear resistance, and strength, making it ideal for soil-engaging tools. The blade is mounted on both ends of the vehicle so that it can simultaneously remove weeds along the watering channels on either side. Since

the blade repeatedly encounters stones, roots, and compact soil, high carbon steel ensures long-lasting performance without chipping or rapid wear, maintaining efficiency during field operation.

Wheel Material – Mild Steel with Polymer Coating

The vehicle uses a pair of permanent wheels made from mild steel for structural support, combined with a polymer outer layer to reduce weight and improve traction. Mild steel provides the strength to support the vehicle load while polymer reduces soil drag and prevents slippage. The rear wheels are detachable and used only when required—such as during ploughing or weed-removal operations—to maintain balance and stability. This combination of materials ensures smooth movement on different field surfaces while keeping the assembly lightweight and easy to maneuver.

Sprayer Tank Material – High-Density Polyethylene (HDPE)

The fertilizer spraying tank is made from HDPE due to its corrosion resistance, chemical stability, and low weight. HDPE is unaffected by fertilizers, pesticides, and water-based chemicals, ensuring that no contamination or structural degradation occurs. The lightweight nature of the tank reduces overall load on the frame, allowing the DC motor to operate efficiently even when the tank is filled. HDPE is also impact-resistant, making it suitable for field conditions where the vehicle may experience shaking or vibrations.

Pump Housing and Nozzle Material – Polymer and Stainless Steel

The fertilizer spraying system uses a small electric pump with polymer housing to reduce weight and resist corrosion. The spray nozzles are made from stainless steel to provide fine, consistent spray patterns while resisting chemical damage

from fertilizers. Stainless steel nozzles maintain their shape even under repeated operation, ensuring uniform fertilizer application across plant rows.

DC Motor and Drive Components – Steel Shaft and Polymer Mounts

The vehicle's motion system is driven by a DC motor selected for its high torque, reliability, and compatibility with battery-powered operation. The motor shaft is made of hardened steel to transmit power without bending, while the mounting brackets are made from polymer to reduce vibration transmission to the frame. This combination ensures stable performance on uneven terrain while keeping the system lightweight.

Battery – 12 Volt Lead-Acid or Lithium-Ion Unit

A 12V battery is used as the primary power source for the DC motor and fertilizer pump. Lead-acid batteries offer affordability and reliability, while lithium-ion options provide lower weight and longer backup. The chosen 12V standard ensures compatibility with agricultural DC motors, relays, and wiring systems. The portable power source allows the vehicle to operate in fields without access to external electricity.

Fasteners, Brackets, and Mounting Hardware – Mild Steel / Stainless Steel

Bolts, nuts, and brackets are selected from either mild steel or stainless steel based on their location in the assembly. Mild steel fasteners are used where high strength is required, while stainless steel is used in the spraying system to avoid corrosion from fertilizer exposure.

CHAPTER 7

WORKING PRINCIPLE

The autonomous fertilizer sprayer operates by integrating the forward movement of a small DC-motor-driven vehicle with the controlled spraying action of a fertilizer pump. The vehicle serves as a mobile platform that travels along crop rows, while the pump disperses liquid fertilizer through nozzles mounted at a fixed height. A microcontroller regulates vehicle motion, pump activation, and spraying duration. Power is supplied by a 12-volt battery, making the system portable and suitable for field use. By coordinating vehicle speed and pump output, the system ensures consistent and uniform application of fertilizer across the plantation without requiring manual spraying.

BRIEF EXPLANATION ABOUT THE PRINCIPLE

The automatic fertilizer sprayer operates on a combined principle of controlled mobility and regulated fluid delivery. The base vehicle moves using DC motors connected to a microcontroller that governs direction and speed. These motors receive pulse-width-modulated signals, allowing the system to adjust acceleration and maintain steady movement across uneven farmland. The microcontroller executes a programmed sequence that controls when the fertilizer pump switches on and off during the vehicle's travel path.

The fertilizer storage tank is connected to an electric pump that draws the liquid and sends it through spray nozzles. As the pump activates, fertilizer is atomized through the nozzles in a fine spray pattern. The spray angle, nozzle spacing, and flow rate are selected to cover plant rows effectively while minimizing wastage. The microcontroller uses timing logic or optional sensor feedback to maintain uniform spray output regardless of minor changes in vehicle speed.

The entire system is powered by a 12V DC battery, ensuring autonomous field operation without depending on external electrical supply. Voltage regulation modules ensure stable power delivery to the DC motors, pump, and control electronics. The use of a battery-based setup makes the sprayer suitable for remote agricultural fields where power access is limited.

The coordinated functioning of mobility and spraying ensures that fertilizer is applied evenly across the crop area. This eliminates inconsistent manual spraying, reduces operator fatigue, and supports precise and efficient agricultural treatment across crops of varying sizes—not limited to cotton plantations but adaptable to vegetables, cereals, orchards, and horticulture crops.

CHAPTER 8

Process Involved in Fabrication

CUTTING

The fabrication process begins with cutting raw materials such as mild-steel tubes, metal plates, brackets, and polymer sheets to the required dimensions. These materials form the structural frame and housing for the vehicle and the fertilizer tank. Cutting is performed using tools such as hacksaws, angle grinders, or power saws for basic operations, while laser cutters or CNC machines are used when higher accuracy or uniform shapes are needed. Proper measurement and marking before cutting ensure that all structural components match the design specifications. Accurate cutting is essential for proper alignment and stability when the frame and mechanical elements are assembled.

WELDING / JOINING

After cutting, the next step is joining the structural components to form the main chassis of the fertilizer spraying vehicle. Welding is used for permanent joints, especially where strength and rigidity are required to support the battery, motor assembly, and fertilizer tank. Bolts and nuts are used where disassembly or adjustments may be necessary, such as mounting the pump, nozzles, electronics, and motor brackets. Maintaining correct alignment during joining prevents frame distortion and ensures that the vehicle moves smoothly during operation. Safety measures like gloves, eye protection, and proper ventilation are followed to prevent injuries.

DRILLING

Drilling is carried out to create mounting points for the DC motors, wheel shafts, pump, battery holder, nozzle brackets, and electronic modules. Holes are drilled using hand drills or bench drilling machines depending on the precision required. Accurate drilling ensures that components are mounted firmly and aligned correctly, which is essential for vibration-free motion and stable spraying. Burrs and rough edges are removed after drilling to prevent misalignment and interference with moving parts. Proper spacing and symmetry of drilled holes contribute to the smooth movement and load distribution of the vehicle.

GRINDING / FINISHING

After joining and drilling, grinding and finishing are done to remove sharp edges, weld splatter, and surface irregularities on the frame. This step improves safety by eliminating sharp corners and enhances the overall appearance of the vehicle. Grinding also smoothens contact points where the fertilizer tank, electronics, and motor housings are placed, ensuring firm and stable mounting. Surface finishing reduces friction between moving parts and prevents unnecessary wear during long-term operation. Finishing also prepares the frame for protective coatings, which help prevent corrosion when the machine is used in outdoor agricultural conditions.

ASSEMBLY

Assembly is the final stage, where all mechanical, electrical, and fluid components are integrated. The steel frame is assembled first, followed by the installation of wheel shafts, DC motors, and the drivetrain. The fertilizer tank is mounted securely, and the electric pump is connected to the spray nozzles through flexible pipes. The 12V battery, wiring harness, and microcontroller unit are installed and connected to control the movement and pump operation. Nozzles

are positioned at the required height and angle for proper fertilizer distribution. The vehicle is tested for alignment, stable motion, and leak-free spraying. Proper fastening and adjustment during assembly ensure smooth movement, efficient pumping, and reliable automatic spraying across various crop fields.

CHAPTER 9

Modeling (2D & 3D Design)

Fig 9.1 2D DESIGN

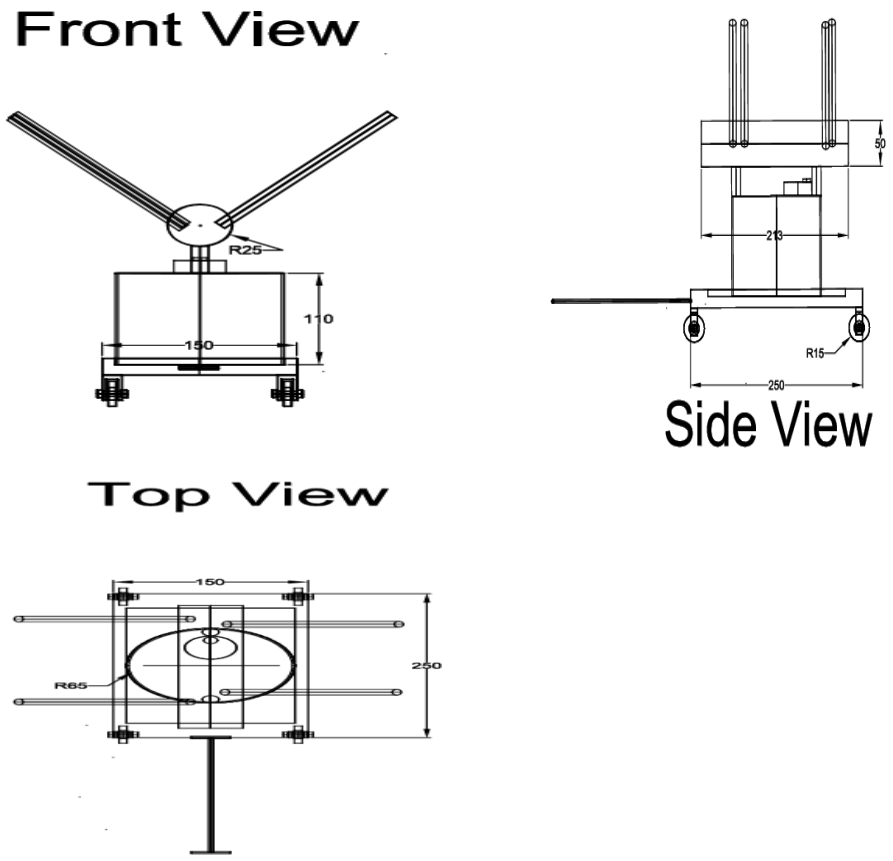
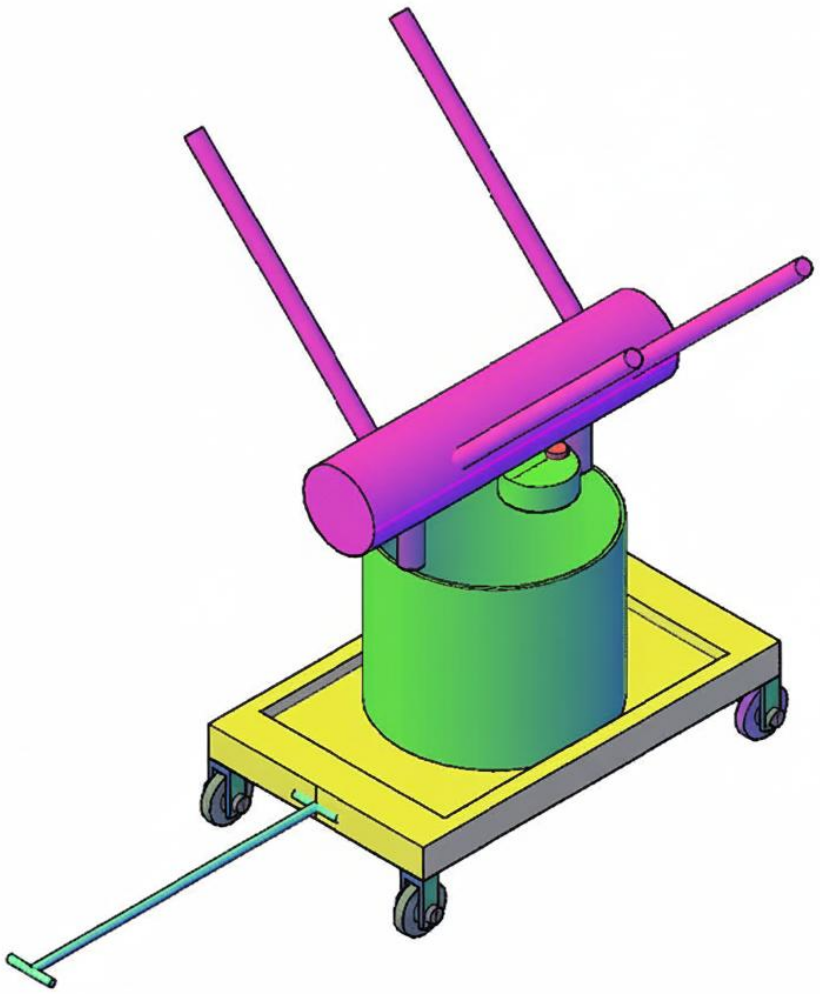


Fig 9.2 3D DESIGN



CHAPTER 10

Budget

. No.	Component	Quantity	Total Cost (₹)
1	Mild Steel Frame Material	1 set	1,200
2	DC Motor	1	900
3	Wheel Assembly	4	1,000
4	Fertilizer Tank	1	350
5	Sprayer Pump	1	450
6	Spray Nozzles & Pipe	1 set	250
7	Plough/Weeding Blade	2	600
8	Battery	1	1,000
9	Switches & Wiring	1 set	250
10	Fasteners & Accessories	1 set	200
11	Fabrication Cost	—	800
Total Project Cost:			**7000**

REFERENCE

1. Prakash, S., et al. (2023). *Design and development of a multipurpose agricultural vehicle for integrated ploughing, seed sowing, and fertilizer spraying*. Journal of Agricultural Machinery and Automation, 12(4), 45–53.
2. Karthik, A., et al. (2022). *Solar-powered multipurpose agricultural machine for ploughing, spraying, and sowing*. International Journal of Renewable Agricultural Systems, 8(2), 101–110.
3. Patel, R., et al. (2021). *Battery-operated weeder and sprayer for small-scale farming applications*. Advances in Farm Mechanization, 6(1), 27–35.
4. Deepak, M., et al. (2023). *Fabrication of an automated multipurpose agricultural robot for ploughing, watering, and spraying*. International Journal of Smart Agriculture Technology, 14(3), 89–98.
5. Ganesh, P., et al. (2020). *Development of a low-cost multipurpose agricultural vehicle for weeding, ploughing, and fertilizer spraying*. Journal of Rural Mechanization and Field Engineering, 5(2), 60–68.